On Canonical Liftings, Group Actions, and Linkable Ring Signatures

Joint work by:
Alessandro Budroni, Giuseppe D'Alconzo, Leonardo Errati, Edoardo Persichetti

Politecnico di Torino
Department of Mathematical Sciences

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Section 1

Signature Compression

Group Actions I

Let $(G, \circ)$ be a group and X a set. A *group action of G on X* is a map

$$\star : G \times X \to X$$

compatible with the group operation:

$$\mathrm{id}_G \star x = x$$
$$g \star (h \star x) = (g \circ h) \star x$$

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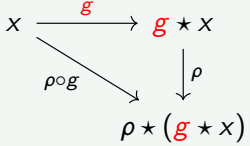


$X =$ possible states of the cube

$G = (\mathsf{Z}_3^7 \times \mathsf{Z}_2^{11}) \rtimes ((\mathsf{A}_8 \times \mathsf{A}_{12}) \rtimes \mathsf{Z}_2).$

Group Actions II

Remark: Sigma Protocol



$$\mathcal{R} = \{\langle (x, y), g \rangle \mid x = g \star y\rangle\}$$

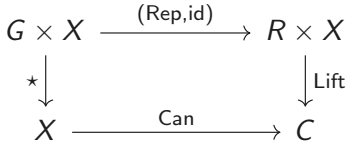
Definition: GAIP

Given $x, y \in X$, find $g \in G$ with $x = g \star y$.

Q1: can we compress responses?

Canonical Liftings I

A *canonical lifting* for a group action $(G, X, \star)$ is a tuple $(R, C, \mathsf{Rep}, \mathsf{Can}, \mathsf{Lift})$ of sets and efficient maps such that this diagram commutes:



$$\mathsf{Rep} : G \to R$$
$$\mathsf{Can} : X \to C$$
$$\mathsf{Lift} : R \times X \to C$$

Definition: Defining Property

$$\mathsf{Can}(g \star x) = \mathsf{Lift}(\mathsf{Rep}(g), x)$$

Canonical Liftings II

A canonical lifting is *effective* if

1. it is efficient to find a g with given representative r
2. if $\mathrm{Can}(x) = \mathrm{Can}(y)$, it is efficient to find g with $y = g \star x$.

Definition: CLCP

Given $x, y \in X$, find $r, s \in R$ with $\mathrm{Lift}(r, x) = \mathrm{Lift}(s, y)$.

If a canonical lifting is *effective*, GAIP and CLCP are PPT-equivalent.
It abstracts / adds to known compression techniques.

Section 2

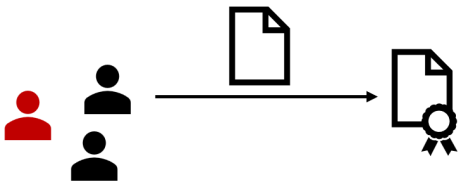
Advanced Signatures

Ring Signatures [RST01]

"*ring*" of public keys $\quad R = \{pk_1, \ldots, pk_n\}$

A *ring signature* allows a member of R to sign on behalf of R without revealing its index.

$$RS = (KeyGen, Sign, Verify)$$

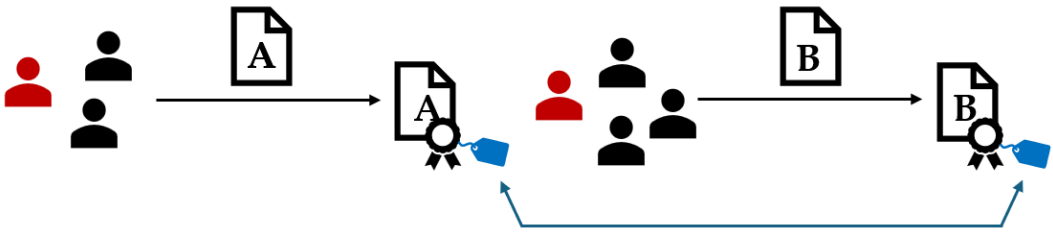


Linkable Ring Signatures [LWW04]

"*ring*" of public keys $\quad R = \{pk_1, \ldots, pk_n\}$

Linkable if it enables to verify whether two signatures were produced by the same sk_i.

$$LRS = (KeyGen, Sign, Verify, Link)$$



Linkable Ring Signatures [LWW04]

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$$\mathsf{LRS} = (\mathsf{KeyGen}, \mathsf{Sign}, \mathsf{Verify}, \mathsf{Link})$$

Security properties:
Anonymity of signer.
Linkability of repeated uses of sk_i.
Non-frameability of an honest user.

Literature varies. We picked the strongest variants [BOA24].

Advanced Signatures via Canonical Liftings

base sigma protocol

↓

signature

security problem: CLCP

slight modification: 5 to 8% faster LESS signing

OR sigma protocol

↓

ring signature

security problem: CLCP

action commutativity is not required

Q2: Is it easy to add linkability?

Advanced Signatures via Canonical Liftings



OR-t sigma protocol

↓

linkable ring signature

security problem:
non-classical

LRS from Group Actions [BKP20]

RS: for a commutative group action, apply Fiat-Shamir to a sigma protocol for

$$\mathcal{R}^{\mathrm{OR}} = \left\{ \langle (x_1, \ldots, x_N, y), w \rangle \in X^{N+1} \times W \mid \text{ for some } i \in [N], (x_i, y, w) \in \mathcal{R} \right\}.$$

LRS: tag-key relation via *witness-reuse*.

$$\mathsf{sk} = g \star \mathsf{pk}, \qquad \mathsf{tg} = g \circ \mathsf{t}$$

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Non-commutative actions may not meet the security hypotheses of LRS [IGL26].

In codes & lattices, choosing $\star = \circ$ leads to key recovery [BCDD+24, BCDF25].

LRS from Canonical Liftings

Idea: tag-key relation is g, g' related by $\mathsf{Rep}(g) = \mathsf{Rep}(g')$.

Definition: Left-compatibility

1. $\mathsf{Rep}(g) = \mathsf{Rep}(h) \implies \mathsf{Rep}(\rho \circ g) = \mathsf{Rep}(\rho \circ h)$
2. given g, sampling uniform h such that $\mathsf{Rep}(g) = \mathsf{Rep}(h)$ is efficient

linkability	*linkable anonimity*	*non-frameability*
non-splitting + left-compatible	**2-pseudorandom** + left-compatible	**frame one-way** + left-compatible

Performance of CELeReS

Scheme	Security	2^1	2^2	2^6	2^{10}	2^{12}	Assumption
CELeReS [BDEP26]	128 bit	26.7	27.8	32.3	36.7	39.0	OW, F-OW, 2-PR
	192 bit	69.2	72.3	85.1	97.8	104.2	
	256 bit	119.1	123.8	142.6	161.3	170.7	
Raptor [LAZ19]	100 bit	–	–	82.7	1301.9	5203.3	NTRU
Calamari [BKP20]	NIST-I	3.6	4.6	14.1	20.1	23.1	CSIDH-512
Falafl [BKP20]	NIST-I	33	34	36	39.3	41	MSIS, MLWE
DualRing-PRF [ZSL+24]	128 bit	–	–	45	64	–	Residue PRFs

Signature size (kB) for varying ring size. Comparison with post-quantum schemes adopting the strongest security definitions for LRS.

CELeReS-o (I)

Tags are large: 16.2 kB (of 26.7 kB) in one case.
One possible compression technique: self-orthogonal tag matrices [BMSS25].

Security	2^1	2^2	2^6	2^{10}	2^{12}
128 bit	18.9	19.9	24.5	28.9	31.2
192 bit	50.8	54.0	66.7	79.5	85.9
256 bit	85.4	90.1	108.8	127.6	136.9

New tags: 8.3 kB (of 18.9 kB) for the same case.

CELeReS-o (II)





faster key generation

smaller tags

Summary

What we showed

- This abstracts compression techniques
- They can build classical and (linkable) ring signatures

Open problems

- Different group actions?
- New techniques/functionalities?

https://eprint.iacr.org/2026/1348

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- Alessandro Budroni, Giuseppe D'Alconzo, Leonardo Errati, and Edoardo Persichetti, *On canonical liftings, group actions, and linkable ring signatures*, Cryptology ePrint Archive, Paper 2026/1348, 2026.

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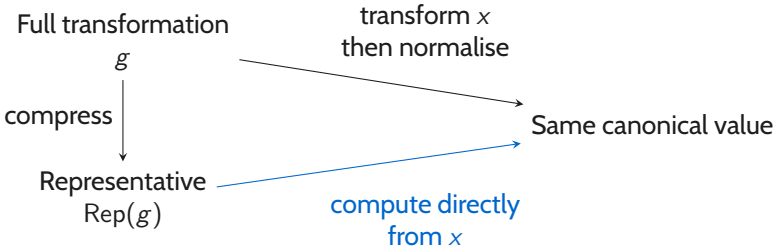


Xinyu Zhang, Ron Steinfeld, Joseph K Liu, Muhammed F Esgin, Dongxi Liu, and Sushmita Ruj, *Dualring-prf: Post-quantum (linkable) ring signatures from legendre and power residue prfs*, Australasian conference on information security and privacy, Springer, 2024, pp. 124–143.

EXTRA SLIDES

open in case of emergency

Canonical Liftings: enchore



Defining Property

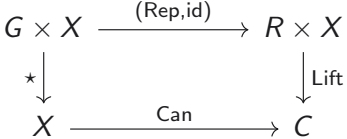
$$\mathrm{Can}(g \star x) = \mathrm{Lift}(\mathrm{Rep}(g), x)$$

Which lifting for CELeReS?

The canonical form used in LESS is not left-compatible.
Instead, we adapt existing constructions.

$$X = \mathbb{F}_q^{k \times n}, \qquad G = \mathrm{Mon}_n(\mathbb{F}_q)$$

$$C = \mathrm{CF}(X)\ [\mathrm{DKQ}^+25], \qquad R = \mathrm{Perm}_n(\mathbb{F}_q)$$



$$\mathrm{Rep} : \boldsymbol{PD} \mapsto \boldsymbol{P}$$
$$\mathrm{Can} : \boldsymbol{M} \mapsto \mathrm{CF}(\boldsymbol{M})$$
$$\mathrm{Lift} : (\boldsymbol{M}, \boldsymbol{P}) \mapsto \mathrm{CF}(\boldsymbol{MP})$$

Security Properties of LRS

linkability

Find α in $\mathrm{MAut}(\boldsymbol{G}) \setminus \mathrm{MAut}(\boldsymbol{T})$.

If $\boldsymbol{G}, \boldsymbol{T}$ are random, they are trivial with overwhelming probability.

linkable anonimity

Distinguish between

$(\mathrm{RREF}(\boldsymbol{GQ}), \mathrm{RREF}(\boldsymbol{TQ}'))$

$(\mathrm{RREF}(\boldsymbol{GQ}_0), \mathrm{RREF}(\boldsymbol{GQ}_1))$

A 2-sample version of 2-pseudorandomness for the quotient action by $\mathrm{Perm}_n(\mathbb{F}_q)$. The best attacks are those on LEP.

non-frameability

(If $\mathrm{MAut}(\boldsymbol{T})$ *is trivial)*
Given $(\boldsymbol{G}, \boldsymbol{G}')$ and $(\boldsymbol{T}, \boldsymbol{T}')$ pairs of linear equivalent codes under the same $\boldsymbol{P}$, find $\boldsymbol{P}$.

The best attacks are those on LEP.

LRS from Canonical Liftings: CELeReS

Code Equivalence Linkable Ring Signature

The lifting of LESS is not left-compatible. Our lifting has $\mathrm{Rep}(\boldsymbol{Q}) = \boldsymbol{P}$.
Idea: $\boldsymbol{Q} = \boldsymbol{PD}, \boldsymbol{Q}' = \boldsymbol{PD}'$ related by $\boldsymbol{P}$

Left-compatibility

1. if $\boldsymbol{Q}, \boldsymbol{Q}'$ share permutation, so do $\boldsymbol{MQ}, \boldsymbol{MQ}'$
2. sampling uniform $\boldsymbol{Q}''$ with same permutation is efficient

linkability

if $\mathrm{Aut}(\boldsymbol{G}), \mathrm{Aut}(\boldsymbol{T}) = \{1\}$, it holds unconditionally

linkable anonimity

best known attacks are those on LEP

non-frameability

best known attacks are those on LEP

Full performance of CELeReS

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Scheme	Ring size	KeyGen	Sign	Verify
CELeReS	2^1	$1.32\,M\ /\ 4.5 \times 10^{-4}\,s$	$421\,M\ /\ 0.141\,s$	$397\,M\ /\ 0.137\,s$
	2^3		$2419\,M\ /\ 0.806\,s$	$2035\,M\ /\ 0.678\,s$
	2^6		$9300\,M\ /\ 3.090\,s$	$7865\,M\ /\ 2.631\,s$

Faster LESS signing

Extra payoff from canonical liftings

Instead of sampling a full ephemeral transformation, sample its representative directly. This requires a well-defined efficient operation

$$r \odot g = \mathrm{Rep}(g_r \circ g), \qquad \mathrm{Rep}(g_r) = r,$$

and sampling representatives with the correct induced distribution.

5 to 8% improvement in LESS signing time.